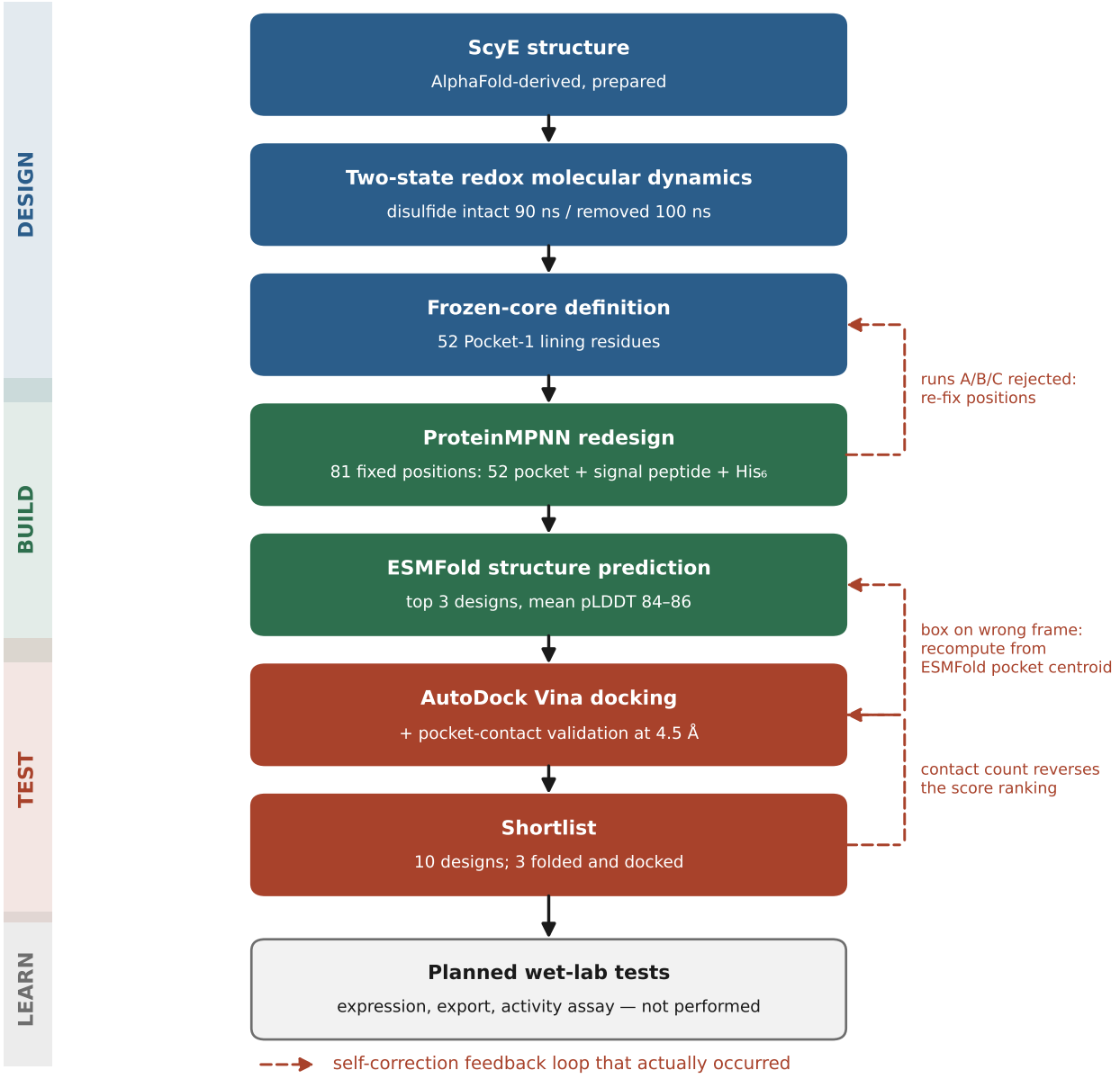
DESIGN
BUILD
TEST
LEARN
ScyE structure
AlphaFold-derived, prepared
Two-state redox molecular dynamics
disulfide intact 90 ns / removed 100 ns
Frozen-core definition
52 Pocket-1 lining residues
ProteinMPNN redesign
81 fixed positions: 52 pocket + signal peptide + His6
ESMFold structure prediction
top 3 designs, mean pLDDT 84–86
AutoDock Vina docking
+ pocket-contact validation at 4.5 Å
Shortlist
10 designs; 3 folded and docked
Planned wet-lab tests
expression, export, activity assay — not performed
runs A/B/C rejected:
re-fix positions
box on wrong frame:
recompute from
ESMFold pocket centroid
contact count reverses
the score ranking
self-correction feedback loop that actually occurred